When Correctness Is the Prediction: An Evaluation Trap in Sampling-Based Uncertainty for Handwritten Mathematics

Anonymous WACV Datasets Track submission

Paper ID *****

Abstract

Repeated sampling provides a cheap, label-free uncertainty signal through model self-disagreement. We evaluate it on handwritten mathematics and find a useful transcription signal, no grading signal, and an evaluation artifact that can manufacture an apparent effect. Against the frozen transcription scorer, entropy reaches AUROC 0.835*, replicated at* 0.828 *on a second model family. It survives parser and ceiling controls and outperforms token confidence and simpler summaries of the same samples. When all five samples disagree, the scorer marks* 57 *of* 61 *items incorrect. For grading, however, entropy is indistinguishable from chance (AUROC* 0.520*).*

Stratifying grading by its ground-truth label appears to recover AUROC 0.801*–*0.854 *across three model families, but this effect is arithmetic. Within a fixed-label stratum, correctness is the same binary column as the model verdict, so uncertainty computed from those verdicts predicts correctness almost by construction; a signal-free biased coin outscores every model evaluated here. Transcription does not undergo this fixed-label collapse. Finally, an exploratory single-reader audit of all* 104 *items the scorer marks incorrect attributes approximately three-quarters to the scoring pipeline rather than to the model. Sampling disagreement can therefore support transcription deferral, but its evaluation must test whether the target construction or scorer creates the apparent signal.*

1. Introduction

Automated marking of handwritten mathematics fails silently. When a model misreads 4.9 as 49, the resulting transcription is as fluent, as confident and as well-formed as a correct one, and nothing in the output distinguishes the two. In a deployment where that output contributes to a grade, determining *when* to route a page to a human reviewer is more valuable than a marginal gain in average accuracy, and unlike accuracy it is directly actionable.

Repeated sampling, combined with a measure of the resulting self-disagreement, is the natural candidate for such a signal. Our output-only score requires no task-specific training, labelled calibration data or access to model weights, and therefore applies to closed models served behind an API. It is motivated by self-consistency and semantic entropy [9, 18]; whether it is effective here remains an empirical question. We address that question on FERMAT [13], a benchmark built from textbook and competitive-exam solutions that are systematically perturbed and then manually transcribed and photographed. Its perturbations include both error-inducing and correctness-preserving changes.

Effectiveness proves to be task-dependent, and, more importantly for evaluation practice, the headline figure is easy to misestimate in either direction. On transcription the association with the frozen scorer survives our parser and ceiling controls. On grading it is absent. Between these two findings lies a methodological trap. Confronted with a model whose grading verdict is heavily biased toward a single answer, the natural response is to stratify by the ground-truth label. Doing so appears to recover a substantial effect that replicates across three independently trained model families. That effect is an artifact: within a stratum in which the true label is constant, "was the model correct" is arithmetically the same column as "what did the model answer", so an uncertainty measure computed from those answers predicts correctness almost by construction. Consequently, a signal-free biased coin outscores every model evaluated here. Notably, none of the usual safeguards detected the problem. The pre-registered threshold did not, since the null distribution clears it; statistical power did not, since both strata were adequately powered; and replication did not, since it reproduced the artifact three times.

A second and more easily overlooked failure underlies both arms. The automatic scorer that assigns correctness is itself unreliable. An exploratory single-reader audit of every item it marked incorrect finds that approximately three-quarters of those verdicts are attributable to the scoring pipeline rather than to the model, while targeted checks of items it marked correct reveal unearned passes in the opposite direction. We report sensitivity bounds under explicit recoding assumptions, not corrected population estimates.

This is an evaluation paper rather than a method paper. We propose no new uncertainty estimator. Instead, we establish where a standard estimator is effective, where it is not, and two distinct mechanisms by which an evaluation of it can yield a confident figure that carries no information.

2. Related Work

Sampling-based uncertainty. Repeated sampling turns a generator into an ensemble, and disagreement among the samples is used as a confidence signal. Self-consistency aggregates sampled chains by majority vote and was introduced primarily as an accuracy-improving decoding method, although its vote distribution can also provide an uncertainty estimate [18]. Semantic entropy makes uncertainty explicit by aggregating likelihoods within bidirectional-entailment clusters and taking the entropy over the resulting meanings [9]. Our estimator keeps the resampling and the entropy but replaces both likelihood weighting and semantic grouping with empirical frequencies over exact-match clusters of an *extracted answer span*. That choice is not an oversight. On these data, an NLI-plus-union-find implementation merged opposed clusters through a single false-positive transitive link and degraded as K grew, so the simpler rule is the one that survives (§6). A separate line asks the model to report its own confidence, either as a probability of being correct [8] or in words; we test the verbalized form as a baseline and it carries no signal on the arm we lead with (§4.4).

Selective prediction and abstention. The decision this signal supports is deferral, not correction, which places the work in selective prediction: a model may abstain, and the question is the risk–coverage trade-off it achieves [5]. We summarize this trade-off with AURC against a random-deferral baseline and additionally report coverage at a target risk. In vision–language reasoning, ReCoVERR reduces unnecessary abstention on A-OKVQA and VQAv2 [16]. More directly for OCR, latent-representation probes have been used to abstain from VLM-generated OCR errors [20], and uncertainty-aware OCR has been trained to mark unreliable spans in degraded documents [6]. Our setting instead asks whether output-only resampling supports selective dense transcription of handwritten mathematics. We score against the page-faithful perturbed target, so "correct" means faithful to what is written on the page, not necessarily mathematically true.

Vision–language models on handwritten and mathematical content. MathVista scores answer accuracy across diverse visual contexts, including natural images, diagrams, charts and scientific figures [12]. Document OCR systems such as Nougat instead map pages to source-derived markup and compare against a reference, while noting both reference artifacts and rendering-equivalent formulae [3]. FERMAT [13], the benchmark used here, begins with textbook and competitive-exam solutions, applies controlled perturbations, and has annotators hand-copy and photograph the resulting pages. It supplies both the original correct solution and the page-faithful perturbed version; about 15% of the perturbations preserve correctness. ErrorRadar evaluates first-error localization and categorization [19], whereas ScratchMath evaluates error-cause explanation and classification on authentic student scratchwork [15]. Neither includes an error-free student-response class, although both include correct reference solutions. Thus all three concern the diagnosis of erroneous or perturbed work, but none benchmarks calibrated uncertainty or selective deferral after visual transcription. On real handwritten mathematics exams, GPT-4o grading remains below the reliability needed for deployment [4]; a separate university study reports substantial answer-extraction difficulty and confidence false-positive rates that leave human verification necessary [11].

Evaluation artifacts. Our central negative result is that a metric can manufacture a signal from nothing, and similar failures have appeared in prior work. Some reported emergent abilities can arise from discontinuous metrics rather than model behaviour [14]. Our case is narrower and, we think, sharper: within a stratum where the true label is constant, correctness becomes the same column as the model's own verdict, so a signal-free biased coin outscores every model we tested (§4.6). The scoring machinery is a second source of artifacts. Process-supervised reward models have been trained to score individual steps of mathematical solutions [10]. Math-specific evaluations of LLM judges also find that they can favour stronger candidate models even when their answers are wrong and can be persuaded to inflate incorrect solutions [7, 17]. We observe a different failure mode in one tested judge: on a perturbed-answer benchmark, it reconstructed the mathematically correct answer and graded against *that* rather than against the reference it was given (§6).

3. Setup

Data. FERMAT [13] contains $2{,}244$ photographed handwritten solutions to middle- and high-school problems. The dataset authors apply controlled perturbations to textbook and competitive-exam solutions, and annotators manually transcribe and photograph the results. Each image contains a complete question and solution. Of the $2{,}244$ pages, 340 (15.2%) have only superficial, correctness-preserving modifications; we call these items error-free. That property is load-bearing for this paper: §4.8 shows the analysis in §4.5 is impossible on datasets that lack it. Unless stated otherwise we use a balanced $n = 300$ sample (150 with an error, 150 without), drawn once with a fixed seed.

Tasks. We evaluate two distinct tasks on the same images. *Perception* asks the model to "explicitly perform OCR on the handwritten text and extract the content in LaTeX format"; it is correct when the extracted final answer matches the released page-faithful `pert_a` field after canonicalisation. *Reasoning* asks it to "analyze the Answer to determine whether there is any error" and emit a binary judgement; it is correct when that judgement matches `has_error`. Neither released field is provided directly as text to the model.

Uncertainty measure. For each item we draw $K = 5$ samples at temperature 0.7, reduce each to a canonical label, and take the Shannon entropy of the resulting cluster distribution, $H = -\sum_i p_i \ln p_i$. $H = 0$ when all samples agree and $H = \ln K$ when all differ. We deliberately use exact-match clustering over an *extracted answer span* rather than semantic clustering of whole derivations; §6 records why.

Models. Qwen2.5-VL at 3B and 7B [2], Pixtral-12B [1], LLaVA-NeXT-7B and InternVL3-8B, comprising four independently trained families, all run in bfloat16.

Statistics. Every AUROC carries a 95% bootstrap interval over 10,000 item-level resamples. Comparisons between conditions use a *paired* bootstrap over the same resampled items rather than checking whether two marginal intervals overlap. Before any run we registered a minimum minority-class size of 30 and, for the stratified reasoning hypothesis, an AUROC threshold of 0.70; verdicts below are assigned by a pure function of those thresholds, not by inspection.

Artifacts. The anonymous supplementary artifact provides the evaluation code and tests, exact prompt hashes, a hash-only manifest for the 300-item sample, and an authenticated reconstruction check against a pinned FERMAT revision. Because FERMAT is gated, the public artifact excludes its page images, released text, and model generations derived from those pages; authorized users can reconstruct and verify the sample without redistributing those fields.

4. Results

4.1. Does self-disagreement predict transcription error?

Yes, and by a margin that does not depend on the model's own confidence being poor. On the balanced $n = 300$ sample, perception entropy predicts transcription error with AUROC 0.835 [0.787, 0.879]. The model's own mean token log-probability reaches only 0.537 [0.469, 0.605], an interval that includes chance, and the paired difference is +0.297 [+0.214, +0.380], resolved. The signal is therefore not recoverable from information the model already exposes.

Against the frozen labels, the corresponding deferral rule marks 57 of the 61 items on which all five transcriptions disagree as wrong (93.4% precision, 31.3% recall). Over the full risk–coverage curve this gives AURC 0.410 against a 0.607 random-deferral baseline (Fig. 1).

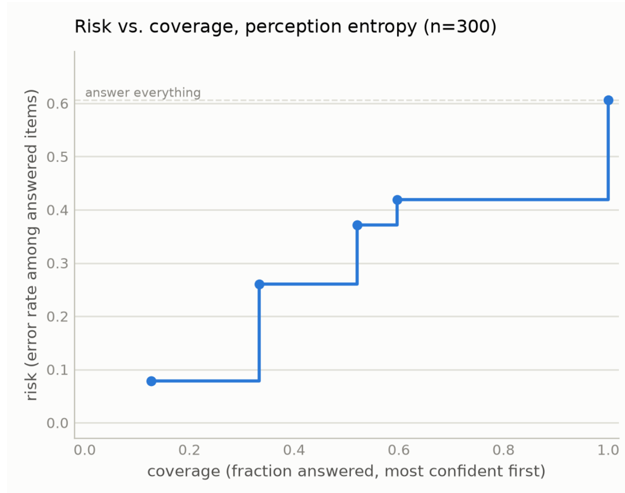

Figure 1. Risk–coverage against the frozen transcription scorer for perception entropy, Qwen2.5-VL-3B, $n = 300$. Deferring the highest-entropy items first traces AURC 0.410 against a 0.607 random-deferral baseline. The curve is the practitioner-facing form of the result: at a 10% risk target the rule retains 21.3% coverage, and token log-probability cannot hold error below 40% at any coverage.

4.2. Does it replicate on a second model family?

Yes. Pixtral-12B, architecturally independent of Qwen, reaches 0.828 [0.782, 0.871] on the same 300 images at a comparable transcription accuracy (41.7% against 39.3%). The intervals overlap along almost their whole length, so the two are not resolvably different.

More important than the point estimate is that it survives the control of §4.4: 0.772 [0.710, 0.833] after removing both parse failures and maximum-entropy items, with 17% of items at the ceiling. §4.9 describes a fourth model that scores *higher* raw and does not survive that control, which is the reason we report the two together.

4.3. What does the entropy summary add?

All five samples per item support several summaries, and a natural objection is that entropy is merely a dressed-up count of distinct answers. Table 1 tests that directly.

Signal	AUROC (95% CI)	vs. entropy
Sampling entropy	0.835 [0.787, 0.879]	n/a
1−majority fraction	0.793 [0.742, 0.841]	+0.042 [+0.011, +0.073]
Distinct count	0.790 [0.739, 0.838]	+0.045 [+0.016, +0.075]
Majority margin	0.779 [0.727, 0.830]	+0.055 [+0.018, +0.092]
Token log-probability	0.537 [0.469, 0.605]	+0.297 [+0.214, +0.380]

Table 1. Perception baselines, Qwen2.5-VL-3B, $n = 300$. The upper block reduces the same five samples in different ways; the lower is a model-internal signal requiring no sampling. Differences are paired bootstraps over the same resampled items, and every one excludes zero. Entropy's margin over the simpler summaries is small but resolved, so the shape of the cluster distribution carries information that a count or a majority does not. The simple summaries are related to entropy but not redundant with it (Spearman 0.83–0.89).

The three sample-based alternatives are computed from the *same* generations at no additional cost, so the comparison isolates the summary rather than the sampling budget.

Two of these deserve comment. The majority margin separates a three–two split from a three–one–one split, which the majority fraction cannot, yet it remains the weakest of the three, because the information entropy exploits resides in the full distribution rather than in the top two clusters. Token log-probability is the only baseline here that is not a function of the samples, and it is also the only one that fails to beat chance; a model that is verbally or numerically confident about a page it has misread is not detectable from its own output distribution, but is detectable from asking it twice. Asking the model directly fares no better. Prompted to state a confidence from 0 to 100 alongside each transcription, it reports a median of 95 while that self-report carries no signal at all: AUROC 0.528 [0.463, 0.593], against 0.807 [0.757, 0.854] for entropy on the same items, a paired difference of +0.279 [0.194, 0.359]. The self-report is not constant, comprising 37 distinct values spanning 68 to 100. The model therefore does vary its stated confidence; it simply varies it without tracking errors, which is the more damaging finding. Two independent cheap-confidence baselines, token log-probability and verbalized confidence, therefore land in the same place, and both are beaten by a resolved margin.

The conclusion is not tied only to five samples. In a matched $K = 10$ run, entropy remains predictive against its frozen label at AUROC 0.806 [0.756, 0.854]. Holding that $K = 10$ label fixed, using all ten samples rather than the first five changes AUROC by +0.045 [+0.005, +0.087] and expands the observed entropy grid from 7 to 34 values. The main practical gain is therefore finer deferral control; this is a sensitivity check, not a post-hoc choice of a better headline K.

Subset	n	AUROC (95% CI)
All items	300	0.835 [0.787, 0.879]
Excl. parse failures	255	0.839 [0.790, 0.886]
Excl. max-entropy items	239	0.794 [0.737, 0.848]
Excl. both	206	0.796 [0.736, 0.852]

Table 2. Perception entropy under artifact controls, Qwen2.5-VL-3B. The signal is graded rather than concentrated at the ceiling: it survives the harshest simultaneous cut with an interval excluding chance. §4.9 shows a model for which it does not.

4.4. Does the signal survive artifact controls?

A cluster-entropy AUROC can be manufactured two ways that have nothing to do with useful uncertainty. Unparseable samples receive their own cluster label, which both inflates entropy and correlates with being scored wrong, so the metric could be measuring the parser. And items pinned at the ceiling $H = \ln K$ can carry the entire ranking if that cell happens to be uniformly incorrect, leaving a binary breakdown detector rather than a graded signal. Table 2 removes each, and both.

Is the reported operating point a tuned one? The sampling temperature and the number of samples are free parameters, so a single setting invites the objection that it was chosen after seeing the result. We regenerated the full transcription arm over a declared grid of three temperatures $\{0.3, 0.7, 1.0\}$ and three sample budgets $\{3, 5, 10\}$, 9,000 fresh generations under a different seed from the frozen run, and applied the controls of Table 2 to each of the nine conditions. The signal is present in all nine: every interval excludes chance after removing both parse failures and ceiling-entropy items. At the reported protocol the regenerated figure is 0.830 [0.784, 0.874], reproducing the locked 0.835 [0.787, 0.879] on independent draws.

The grid also supplies a caution against reading raw numbers across protocols. Uncontrolled AUROC increases with temperature, reaching 0.892 [0.857, 0.924] at $T{=}1.0, K{=}5$, which would appear to argue for hotter sampling. It does not survive the controls: the same condition falls to 0.722 [0.621, 0.815], against 0.741 [0.684, 0.795] at $T{=}0.3$, and the two intervals overlap. The correction grows with temperature because the artifacts do; the parse-failure rate rises from 0.5% to 12.7%, transcription accuracy falls, and at $T{=}1.0$ the controls remove roughly two thirds of the items. The apparent gain is the artifact the controls exist to remove, not a better uncertainty estimate, and we therefore report no temperature effect.

4.5. Does the same signal work for grading?

No. On the same 300 images, reasoning entropy predicts grading error with AUROC 0.520 [0.458, 0.582], an interval containing chance. The balanced sample matters: the model answers "there is an error" for the large majority of inputs, and on an unbalanced set that bias alone would score well.

This is the honest reasoning result, and §4.6 is an account of why we did not initially believe it.

4.6. The trap: stratifying by the label being predicted

A model with a strong answer bias makes a pooled statistic hard to read, and the natural response is to stratify by ground truth and analyse each label separately. Doing so appears to recover a large effect from the null above. Within the `has_error`=1 stratum, entropy predicts grading error at 0.801 [0.751, 0.846]; within the `has_error`=0 (correctness-preserving) stratum it predicts in the opposite direction at 0.280 [0.200, 0.366]. Both strata are adequately powered, both intervals exclude chance, the sign reversal has an appealing mechanistic story, and the effect reproduces across three independently trained model families (Table 3). We reported this as our main finding.

It is an artifact, for a reason that applies to any binary-decision task evaluated this way. **Within a stratum where every item shares the same true label, "was the model correct" is not a separate fact from "what did the model answer"; the two are the same column.** On our 7B run they agree on 100% of items, and the AUROC against correctness is numerically identical to the AUROC against the model's own verdict. Entropy is computed from precisely the votes that produce that verdict, so the measurement is near-circular. The sign reversal follows by the same identity: in the `has_error`=0 stratum correctness is the exact *complement* of the verdict, so anything correlating with the verdict must invert. The two stratum AUROCs sum to 1.

The effect is not specific to the frozen $K = 5$ protocol. A grid over $K \in \{3, 5, 7, 9, 15\}$ and response bias $p = 0.05, \ldots, 0.95$ shows the same pattern: the artifact vanishes at $p = 0.5$, grows as the response bias increases, and is removed by balanced pooling.

The mechanism is arithmetic rather than behavioural. On the error-containing response subset, the majority vote is wrong only when most samples differ from the model's default, and those minority votes are exactly what raise entropy. Nothing about the model's competence is required for the association to appear.

Three things would have caught this earlier, and none of them is specialised. Comparing against a signal-free null rather than against chance, since 0.5 is the wrong reference point for a stratified statistic. Checking whether the correctness label is a relabelling of the prediction before computing anything. And treating a pre-registered threshold as necessary but not sufficient, since ours was cleared by the null.

Model	Observed	Bias-only null	Collapse
Qwen2.5-VL-3B	0.854	0.901 [0.849, 0.932]	99.8%
Qwen2.5-VL-7B	0.801	0.868 [0.821, 0.903]	100.0%
LLaVA-NeXT-7B	0.775	0.831 [0.770, 0.874]	97.2%

Table 3. The apparent replication, against a null with no item-level signal at all: a model answering "error" with fixed probability per sample, so its entropy is pure sampling noise. The null *exceeds* every observed value. "Collapse" is the fraction of items on which correctness and the model's own verdict are the same label. Even the null's 2.5th percentile clears the 0.70 threshold we had registered in advance as confirming the hypothesis, so that threshold could never have separated signal from arithmetic.

4.7. Why transcription is not affected

Both tasks have ground truth; what differs is how many values it can take. Grading truth is binary, so stratifying by it makes the truth *constant* within each group, and correctness reduces to correct = (majority vote = that constant), a function of the vote, which is precisely what entropy summarises. Transcription truth is a mathematical expression drawn from an effectively unbounded set. Five samples can agree perfectly on $\frac{5}{2(x-1)}$ and still be wrong because the reference was $\frac{5}{2(x+1)}$, so agreement carries no guarantee of correctness. In the data, 38 items are unanimous and 3 of those are wrong, while 4 items at maximum entropy are right; correct and incorrect items occur at both ends of the entropy range.

It is worth being precise about which ingredient does the damage. The collapse is caused by *stratifying* rather than by the label type; grouping transcription items by their exact reference answer would collapse in exactly the same manner. Low cardinality is what makes stratification both tempting and total: with a binary label it is the natural response to class imbalance, and it yields precisely two single-label groups. With hundreds of distinct answers nobody would stratify, because the groups would have one member each. The trap is therefore specific to binary-decision tasks, which is where it is most likely to be walked into.

Consistent with this, §4.3 shows transcription entropy outperforming simpler summaries of the same votes; a measure that were merely counting minority votes could not.

4.8. What happens on a second dataset?

The analysis in §4.5 requires both error-containing and error-free evaluated solutions. ErrorRadar [19] supplies correct and incorrect solutions as paired fields, but every evaluated candidate solution contains an annotated error. ScratchMath [15] includes correct answer and solution fields, but every evaluated student answer is labelled incorrect. Neither therefore supplies an error-free response stratum for our binary grading protocol, so neither can support a balanced replication or exhibit the inversion.

On ScratchMath, which is the only one of the two with handwriting to grade, the measurement fails for a second and more instructive reason. Grading accuracy reads 96.0%, but every evaluated student answer is erroneous and the model answers "error" for 90% of them, so it scores well by construction. In 24% of readings $(120/500)$ the model explicitly states it cannot read the image, and in 82% of those $(98/120)$ it answers "error" regardless. Excluding those readings does not clean the estimate: all four misgraded items lie among them, so the retained subset has no minority class and the AUROC is undefined rather than improved. We therefore report this setting as *gated*: not a negative result about uncertainty, but a statement that the pair cannot test it.

4.9. When does a high score mean nothing?

InternVL3-8B attains the highest raw perception AUROC in our study, 0.915 $[0.880, 0.944]$, on a transcription accuracy of 21.0%, approximately half that of the model in §4.1. Independently recomputing every derived column from raw output reproduces the stored values exactly, so this is model behaviour rather than a scoring defect. Under the control of §4.4 it collapses to 0.556 $[0.495, 0.614]$.

The mechanism is visible in the entropy distribution: 202 of 300 items sit at the ceiling, and accuracy within that group is 0.99% against 62.2% elsewhere. The score is separating total generation breakdown from everything else, which is a binary property of the output, not graded uncertainty about content. Reported without the control, it would have been the strongest number in this paper.

5. Failure Analysis

For an exploratory failure analysis, one reader inspected and coded every item the frozen rule marked wrong on Qwen2.5-VL-3B, a complete census of that bucket comprising 104 of 104 items, working from the handwritten pages rather than from the extracted text. The distribution is not what the aggregate number suggests:

coded reason	n	share
scoring/extraction attributable	77	74.0%
notation misread	22	21.2%
undecidable from the page	4	3.8%
copied a different line	1	1.0%
hallucination	**0**	**0%**

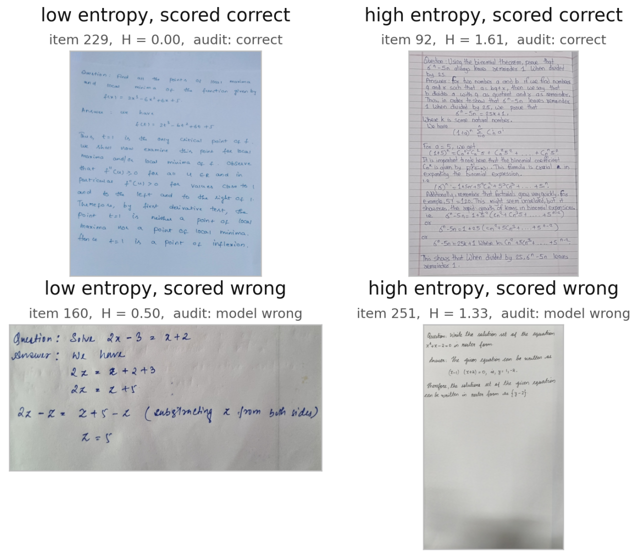

Figure 2. Four perception cases. Self-disagreement helps when wrong answers produce varied samples, but it is weakest when the samples mostly agree on the same wrong reading. The top panels were scored correct and the bottom panels were scored wrong; the left panels have low entropy and the right panels have high entropy. Each panel gives its item, entropy and human-audit verdict.

Three things follow. First, the reader assigned **zero hallucination labels in this 104-item bucket**; this is evidence about the audited set, not a population-level claim that the model never hallucinates. Second, roughly three-quarters of what the scorer calls model error is attributable to the scoring pipeline, either through the extractor selecting a wrong or partial span, or through normalization collapsing a long answer onto a single symbol. Third, and this is the constraint on the method rather than on the model, *the errors that remain are not the ones entropy can flag*: a low-entropy wrong item is one where the samples mostly agree, so self-disagreement has little to use, whichever way the label happens to fall (Fig. 2, bottom-left panel).

Two mechanisms recur in the coded notes and both concern the final answer statement rather than the mathematics. The model verifies a solution's core computation and stops, without checking whether the final statement is complete (a variable never substituted), internally consistent (a tuple introducing z while the condition repeats x), or licensed by the problem (a unit the problem never gave). And it pattern-matches a stated result instead of recomputing it, accepting $19 \times -18 \times 23 = -7966$ where the correct value is -7866.

The coding is a single reader's, which is its main weakness. We have no second-rater agreement. We do have an intra-rater measurement: 78 items were coded twice in independent passes with different sheets; 47 received determinate labels in both passes, and 4 of those pointed opposite ways.

6. Limitations

The positive result rests on one dataset. It now holds on two model families (§4.2), but two further models could not test it: one sits at a transcription floor of 0.8%, and one fails the artifact control of §4.4. Two of four attempts producing a usable measurement is itself worth stating, since the result requires a model with genuine dense-OCR competence, which is not the common case among open-weight VLMs. Of the two alternative datasets, neither supplies error-free evaluated responses, so neither admits a balanced error-presence sample. Moreover, by §4.6, a `has_error`=1 response set constitutes a single-label stratum, on which the measurement collapses for the same reason.

We report the correction to §4.6 as a finding, but it was found late. It was not caught by the pre-registration, which set a threshold the null clears; nor by adequate power, which both strata had; nor by replication, which reproduced the artifact three times. Cross-model agreement is worth less than it appears when the mechanism is arithmetic rather than behavioural.

Our clustering is exact-match over an extracted answer span, not semantic. Semantic clustering was attempted and abandoned: an NLI-plus-union-find implementation merges opposed clusters through a single false-positive transitive link and degrades at larger K. The reported results avoid that path rather than depending on it being fixed.

Confidence intervals were added after the experiments were designed and run. They corrected several conclusions but cannot supply power the studies were not sized for. $K = 5$ also gives few operating points, which is why two-thirds of conformal calibration splits cannot reach a 30% risk target.

The correctness label is noisy in both directions. Every AUROC above is measured against a frozen string-matching rule, and a single-reader audit indicates that rule has substantial label noise. The census of §5 finds unearned *failures*; a seeded random audit of the items the rule marked *correct* finds unearned passes as well. Under alternative recoding assumptions applied to these audits, transcription accuracy ranges from roughly 44–61%, and AUROC from 0.57–0.73, depending in part on whether an unearned pass is treated as a model error or as undecidable. These are *sensitivity bounds*, not corrected population estimates: the audit sets overlap, two are targeted in one direction, and all coding is from one reader. Table 4 summarizes the audited sets and their sampling restrictions.

This does not erase the frozen-scorer association, and it is worth being precise about what it does. The distinct-answers relationship and the cross-family replication rest on the frozen rule's stored columns, which are unchanged. What is now in question is how well that rule's labels track human judgement. The honest statement is that **entropy predicts frozen-scorer failures; predicting human truth is harder**, and that the automatic scorer is least reliable exactly where answers are long, multi-part, set-valued, or stated as prose.

audit set	n	unearned	caveat
rule said wrong	104	77	complete census; one coder
rule said correct	100	4–16	two passes disagree
flagged tier	108	39	high-risk by construction

Table 4. Exploratory single-reader audit of the frozen scorer. **These rows must not be summed or pooled.** 78 items are shared between sets (union 234, not 312), and two of the three sets can only find errors in one direction by construction, so a pooled agreement figure would be largely tautological. The correct-side range spans two independent passes over disjoint random draws which disagree at $p = 0.00007$. That disagreement, rather than either endpoint, is why accuracy is reported only as a sensitivity range.

No automatic scorer we tried was reliable on our probes, including judges. We evaluated four alternatives to the frozen rule. The frozen rule is brittle in the ways above. A span-primary variant that demotes symbolic normalization to a warning does not eliminate the unearned passes, because where both sides extract the same incorrect span, span matching agrees exactly as the normalizer did; its risk flags do, however, concentrate them. A symbolic equivalence checker is unsafe by construction on this benchmark, because the injected errors are often units, coefficients or notation, exactly what an equivalence checker is designed to erase. Two open math judges failed differently: one rejected a valid probe outright, and the other returned the correct verdict on both probes while its own justification showed it had reconstructed a mathematically correct reference from the problem text and graded against *that* rather than against the perturbed reference it was given, misattributing the reference's answer to the model in the process. That last case is the more instructive: **a verdict-only check cannot detect it**, and on a judge that emits no reasoning it would have been recorded as a clean pass. Grading a perturbed-answer benchmark asks a judge to compare rather than to solve; in our probes, one judge instead attempted to reconstruct and solve the problem.

7. Conclusion

On handwritten transcription, self-disagreement across repeated samples predicts failures of the frozen scorer: it survives the parser and ceiling controls, beats the model's own confidence and simpler summaries of the same samples, and marks 57 of 61 maximum-entropy items as wrong under that scorer. The single-reader audit shows why this must not be equated with human truth. On grading, the same measure carries no signal at all.

The distance between those two sentences is where the work is. Stratifying the grading task by ground truth, an ordinary response to a biased model, produces an effect of 0.80–0.85 that is adequately powered, resolves against chance, clears a pre-registered threshold, and replicates across three independently trained model families. It is still an artifact, because within a single-label stratum the correctness label is the model's own answer under another name. A signal-free biased coin scores higher.

We therefore recommend, for any evaluation of sampling-based uncertainty on a binary decision task: check whether the correctness label is a relabelling of the prediction before computing anything, and compare against a signal-free null rather than against 0.5. Neither is expensive. Together they expose failures that aggregate AUROC can hide.

A third caution follows from the audit in §6 and applies more broadly than to uncertainty work. On a benchmark whose page-faithful answers may be deliberately perturbed, the scorer is part of the experiment. In our tests, the frozen string matcher, a symbolic equivalence checker and two open math judges failed in different ways: one judge rejected a valid probe, while another reconstructed a mathematically correct answer instead of comparing against the supplied reference. Where a correctness label is itself estimated, it deserves the same careful validation as the quantity being estimated against it.

References

[1] Pravesh Agrawal, Szymon Antoniak, Emma Bou Hanna, Baptiste Bout, Devendra Chaplot, Jessica Chudnovsky, Diogo Costa, Baudouin De Monicault, Saurabh Garg, Theophile Gervet, et al. Pixtral 12B. *arXiv preprint arXiv:2410.07073*, 2024. 3

[2] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025. 3

[3] Lukas Blecher, Guillem Cucurull, Thomas Scialom, and Robert Stojnic. Nougat: Neural optical understanding for academic documents. In *The Twelfth International Conference on Learning Representations*, 2024. 2

[4] Adriana Caraeni, Alexander Scarlatos, and Andrew Lan. Evaluating GPT-4 at grading handwritten solutions in math exams. In *Companion Proceedings of the 15th International Conference on Learning Analytics & Knowledge (LAK25)*, pages 126–128, 2025. 2

[5] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, pages 4878–4887. Curran Associates, Inc., 2017. 2

[6] Shuhao Guan, Moule Lin, Cheng Xu, Jinman Zhao, and Derek Greene. Teaching VLMs to admit uncertainty in OCR from lossy visual inputs. In *The Fourteenth International Conference on Learning Representations*, 2026. 2

[7] Yerin Hwang, Dongryeol Lee, Taegwan Kang, Yongil Kim, and Kyomin Jung. Can you trick the grader? adversarial persuasion of LLM judges. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 14632–14651, Suzhou, China, 2025. Association for Computational Linguistics. 2

[8] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022. 2

[9] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2

[10] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *The Twelfth International Conference on Learning Representations*, 2024. 2

[11] Tianyi Liu, Julia Chatain, Laura Kobel-Keller, Gerd Kortemeyer, Thomas Willwacher, and Mrinmaya Sachan. AI-assisted automated short answer grading of handwritten university-level mathematics exam. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 45(1):84–105, 2026. 2

[12] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. MathVista: Evaluating mathematical reasoning of foundation models in visual contexts. In *International Conference on Learning Representations (ICLR)*, 2024. 2

[13] Oikantik Nath, Hanani Bathina, Mohammed Safi Ur Rahman Khan, and Mitesh M. Khapra. Can vision-language models evaluate handwritten math? In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14784–14814, Vienna, Austria, 2025. Association for Computational Linguistics. 1, 2

[14] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems*, pages 55565–55581. Curran Associates, Inc., 2023. 2

[15] Dingjie Song, Tianlong Xu, Yi-Fan Zhang, Hang Li, Zhiling Yan, Xing Fan, Haoyang Li, Lichao Sun, and Qingsong Wen. Can MLLMs read students' minds? unpacking multimodal error analysis in handwritten math. In *Artificial Intelligence in Education*, pages 110–124, Cham, 2027. Springer. Proceedings of AIED 2026; first published online 27 June 2026. 2, 6

[16] Tejas Srinivasan, Jack Hessel, Tanmay Gupta, Bill Yuchen Lin, Yejin Choi, Jesse Thomason, and Khyathi Chandu. Selective "selective prediction": Reducing unnecessary abstention in vision-language reasoning. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages

12935–12948, Bangkok, Thailand, 2024. Association for Computational Linguistics. 2

[17] Andreas Stephan, Dawei Zhu, Matthias Aßenmacher, Xiaoyu Shen, and Benjamin Roth. From calculation to adjudication: Examining LLM judges on mathematical reasoning tasks. In *Proceedings of the Fourth Workshop on Generation, Evaluation and Metrics (GEM^2)*, pages 759–773, Vienna, Austria and virtual meeting, 2025. Association for Computational Linguistics. 2

[18] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2

[19] Yibo Yan, Shen Wang, Jiahao Huo, Hang Li, Boyan Li, Jiamin Su, Xiong Gao, YiFan Zhang, Tianlong Xu, Zhendong Chu, Aoxiao Zhong, Kun Wang, Hui Xiong, Philip S. Yu, Xuming Hu, and Qingsong Wen. ErrorRadar: Benchmarking complex mathematical reasoning of multimodal large language models via error detection. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 24297–24334, San Diego, California, United States, 2026. Association for Computational Linguistics. 2, 5

[20] Jihan Yao, Achin Kulshrestha, Nathalie Rauschmayr, Reed Roberts, Banghua Zhu, Yulia Tsvetkov, and Federico Tombari. Reading between the lines: Abstaining from VLM-generated OCR errors via latent representation probes. *arXiv preprint arXiv:2511.19806*, 2025. 2